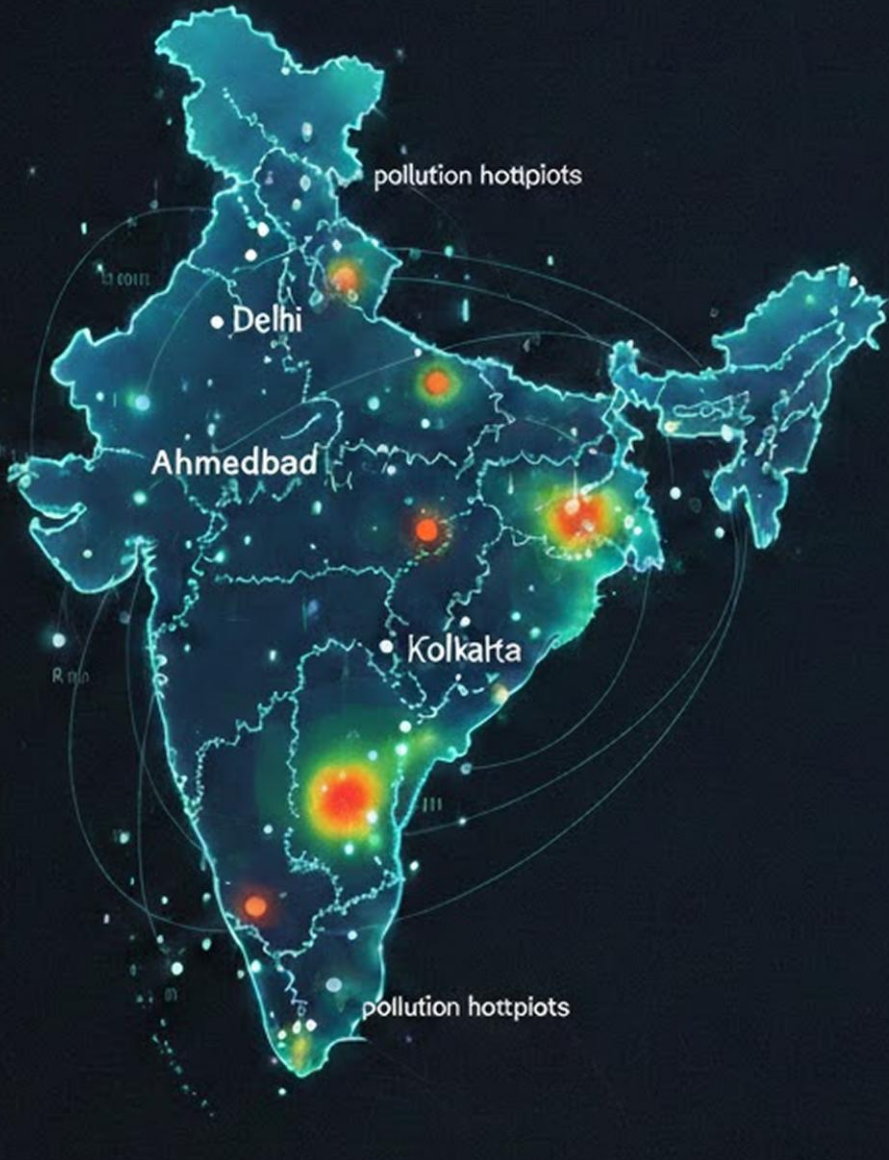




Multidimensional Air Quality Analysis of India: From Data to Policy

Project overview

Data selection strategy

We had four datasets available having different granularity:

- City-level (Daily / Hourly)
- Station-level (Daily / Hourly)

Our Choice: Station-Level Hourly Data (The highest granularity).

Why Station? City averages hide localized hotspots (e.g., Industrial zones vs. Parks).

Why Hourly? Essential to capture intra-day patterns like morning rush hours or night-time emissions.

Project Objectives

1. Analytical Objective: Move from generic pollution levels to specific source identification (e.g., distinguishing Traffic vs. Industrial profiles).
2. Infrastructural Objective: Identify "Blind Spots" where data is insufficient and propose a scientifically optimized expansion of the sensor network.
3. Strategic Objective: Suggest targeted interventions based on the specific pollution profile of each region.

Agenda: From Data to Policy

The project aims to transform raw environmental data into an actionable policy roadmap for India. By combining forecasting, causal clustering, and spatial optimization, we provide a comprehensive framework to understand not just how polluted the air is, but why it is polluted and where we need better data to make informed decisions drove air quality.

Data & Context	Phase I: Forecasting	Phase II: Causal Clustering	Phase III: Uncertainty Analysis
- The Dataset (Indian Air Quality 2015-2020) - Preprocessing and ensuring data integrity (handling missing values, data cleaning) - Exploratory Data Analysis (correlation analysis, distribution check, dynamics analysis)	Workflow - Extending the timeline to ensure data continuity - Bridging gaps using Time Series Modeling (SARIMA & LightGBM & Deep Learning) - Comparing Models results using regression metrics	Workflow - Aggregation and PCA for dimensionality reduction. - Selecting optimal k and applying K-Means clustering - Using Decision Trees to define clear classification rules - Analyzing city pollution structures based on station clusters Goal: Define pollution drivers across different cities of India	Workflow - Using Ordinary Kriging to interpolate AQI across India - Quantifying spatial uncertainty to identify reliability gaps - Proposing sensor locations to maximize network coverage confidence Goal: Define blind spots, where confidence for our data drops

Data description

- Origin:** Central Pollution Control Board (CPCB) official records (via Kaggle).
- Timeline:** 2015 – 2020 (5-year historical window).
- Scope:** Multi-city network covering diverse geographic zones (North, South, East, West).

AQI

PM2.5

PM10

SO₂NO_xNH₃

CO

O₃

Other pollutions

NO

NO₂

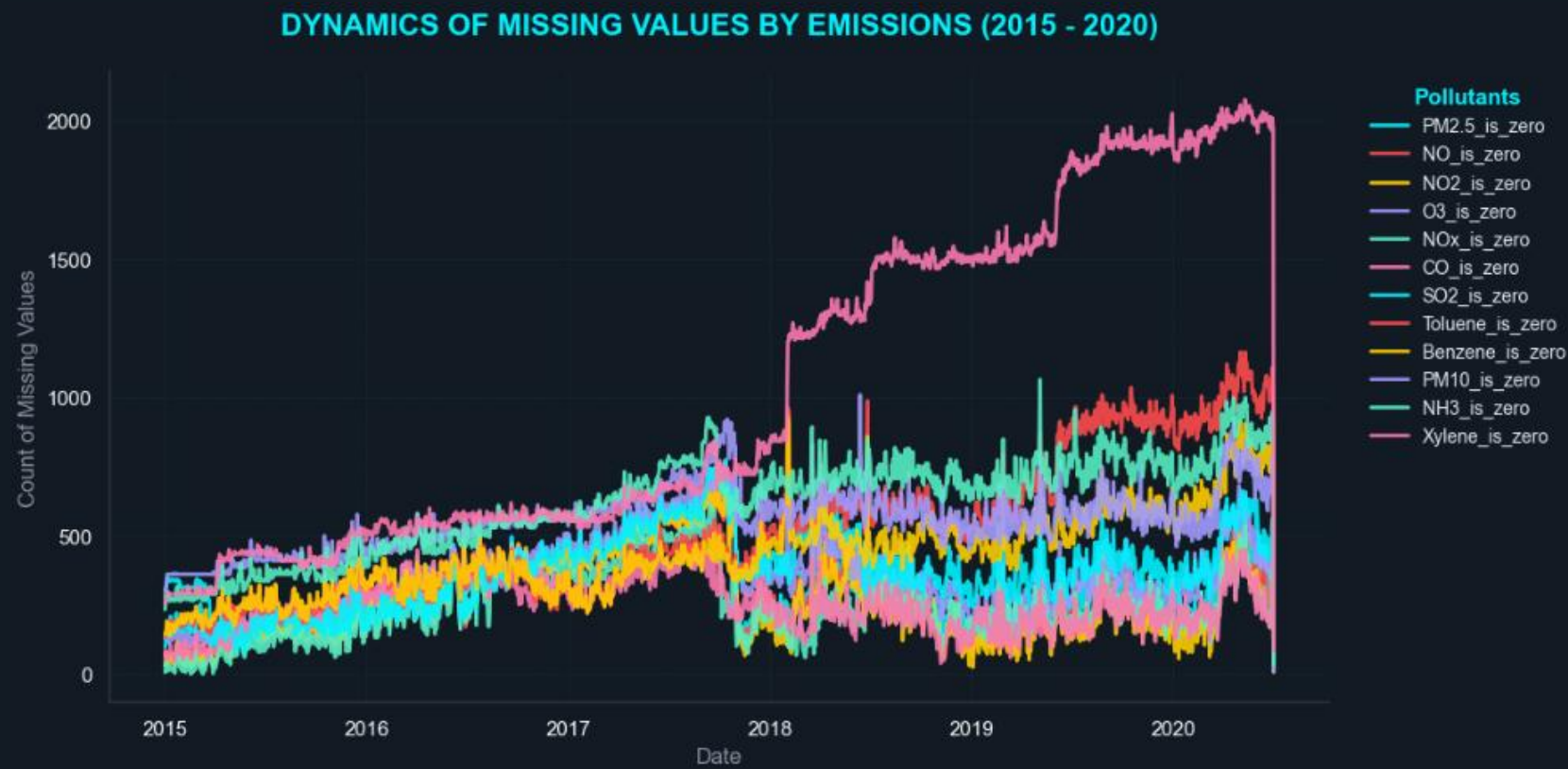
Benzene

Toulene

Xylene

Data preprocessing: missing values

Station_hour dataset contains the information tracking pollutions for each hour for the period from 2015-2020.



	Missing Values	% of Total Values
Xylene	2075104	80.1%
NH3	1236618	47.8%
PM10	1119252	43.2%
Toluene	1042366	40.3%
Benzene	861579	33.3%
SO2	742737	28.7%
O3	725973	28.0%
PM2.5	647689	25.0%
AQI_Bucket	570190	22.0%
AQI	570190	22.0%
NO	553711	21.4%
NO2	528973	20.4%
CO	499302	19.3%
NOx	490808	19.0%

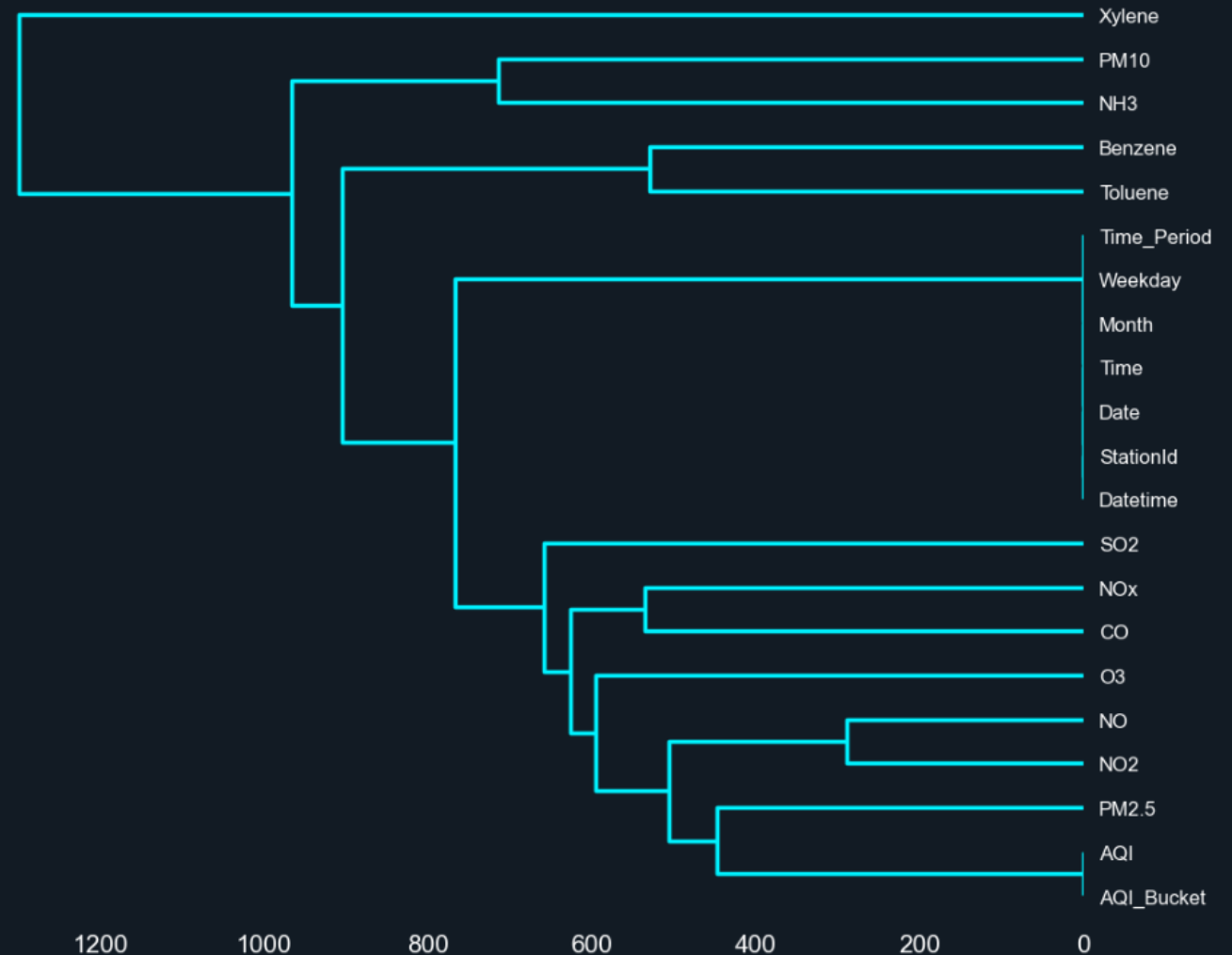
In general, number of missing values has increased significantly in 2018, primarily because of the Xylene
However, there is a positive trend in number of missing values increase over time

Data preprocessing: visualizing gaps

Visualizing Gaps Tool: Utilized the missingno library to generate a Nullity Matrix.

- The variable Xylene exhibits extreme sparsity (>80% missing).
- Horizontal white bands across all variables indicate systemic station failures or power outages.

HIERARCHICAL CLUSTERING OF MISSING VALUES



Data preprocessing: missing values imputation strategy

Our strategy: Fill in missing values for pollutants using Cascading Imputational strategy prioritizing Time-based interpolation and then using AQI formula fill in missing values for the index.

1st layer: **Linear & Time Interpolation** – filling in short gaps

2nd layer: **The Rhythm (Historical Lags)**

If the gap is too large for interpolation, we look at the station's history.

1. Daily Cycle (): We fill missing values with the data from the same hour, yesterday.
2. Weekly Cycle (): If yesterday is missing, we use the data from the same hour, last week.
3. Monthly Hour Mean: If specific history is missing, we use the station's average for that specific hour in that specific month.

3rd layer: **Spatial - City/State imputation**

A station likely behaves similarly to other stations in the same City or State.

4th layer: **Global Median imputation**

Ensures the dataset has 0 missing values for the forecasting models.



1. TEMPORAL INTERPOTATION

Linear & Time-based fills short gaps



2. HISTORICAL RHYTMS

Daily, Weekly, Monthly lags



3. SPATIAL CONTEXT

Neighboring stations (City/State)



4. GLOBAL MEDIAN

Faflwark for baseline continuity

Data preprocessing: missing values imputation strategy

The variable Xylene contains 80% of missing values, therefore it won't be accurate to impute missing values in this case. Moreover, this pollutant is irrelevant and do not participate in the AQI calculation. Thus, we can drop this variable to continue working with data without creating noise.

AQI calculation methodology

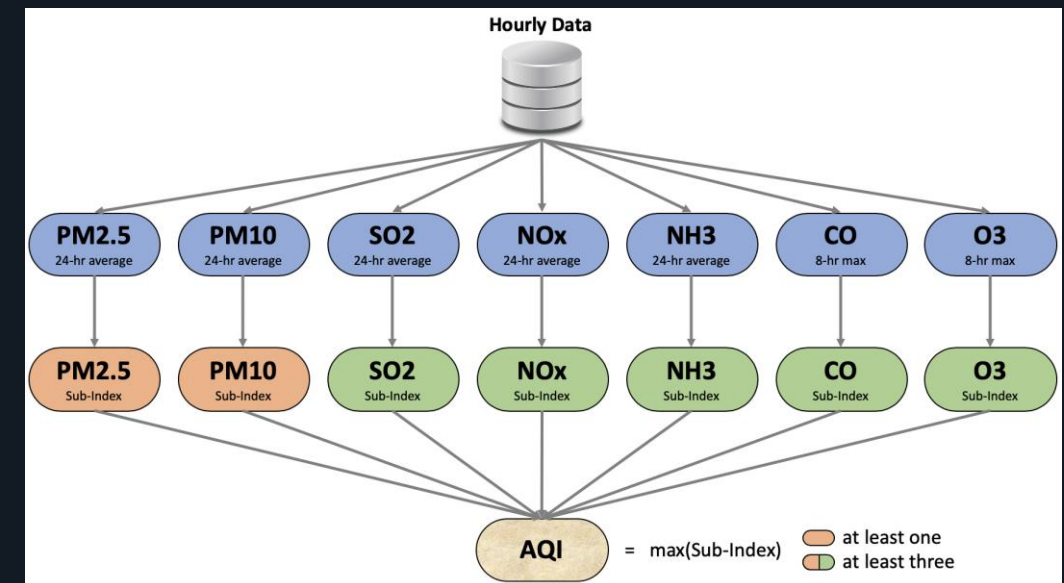
- The AQI calculation uses 7 measures: PM2.5, PM10, SO2, NOx, NH3, CO and O3.
- For PM2.5, PM10, SO2, NOx and NH3 the average value in last 24-hrs is used with the condition of having at least 16 values.
- For CO and O3 the maximum value in last 8-hrs is used.
- Final AQI is the maximum Sub-Index with the condition that at least one of PM2.5 and PM10 should be available and at least three out of the seven should be available

AQI calculation validation

Calculate regression metrics for those variables that satisfy following conditions:

- PM2.5 or PM10 data is available.
- Data exists for at least 3 out of 7 pollutants.
- The station originally calculated the AQI (Target exists).

	Missing Values	% of Total Values
AQI	570190	22.0%
AQI_Bucket	570190	22.0%



Number of objects for validation: 1964352
 R2 Score (accuracy of formula calculation): 0.9539
 MAE (mean absolute error): 6.52

Exploratory Data Analysis

Exploring the correlation between AQI and single pollutants

Impact of Emissions on AQI (sorted by absolute correlation):

PM2.5: 0.654746
PM10: 0.610608
CO: 0.414027
NO2: 0.394572
NOx: 0.355154
NO: 0.320022
SO2: 0.303530
Toluene: 0.271427
NH3: 0.244769
O3: 0.141335
Benzene: 0.106760
Name: AQI_final, dtype: float64

The emissions with significant impact on AQI (corr > 0.6) are PM10 and PM2.5. These results are in alignment with the calculation formula for AQI which takes into special consideration these 2 emissions.
In the calculation methodic we find condition that PM2.5 and PM10 have to be present to calculate AQI

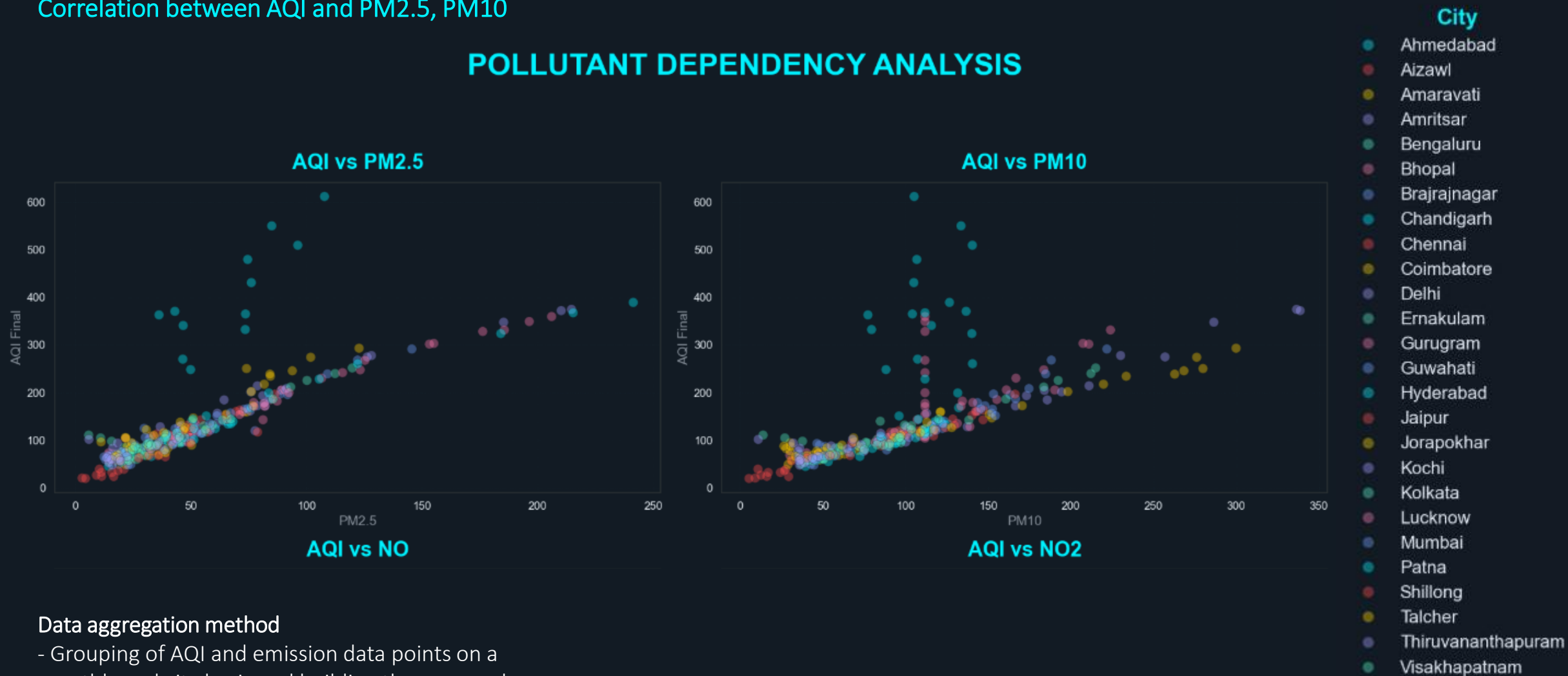
CORRELATION: EMISSIONS & AIR QUALITY INDEX



Exploratory Data Analysis

Correlation between AQI and PM2.5, PM10

POLLUTANT DEPENDENCY ANALYSIS



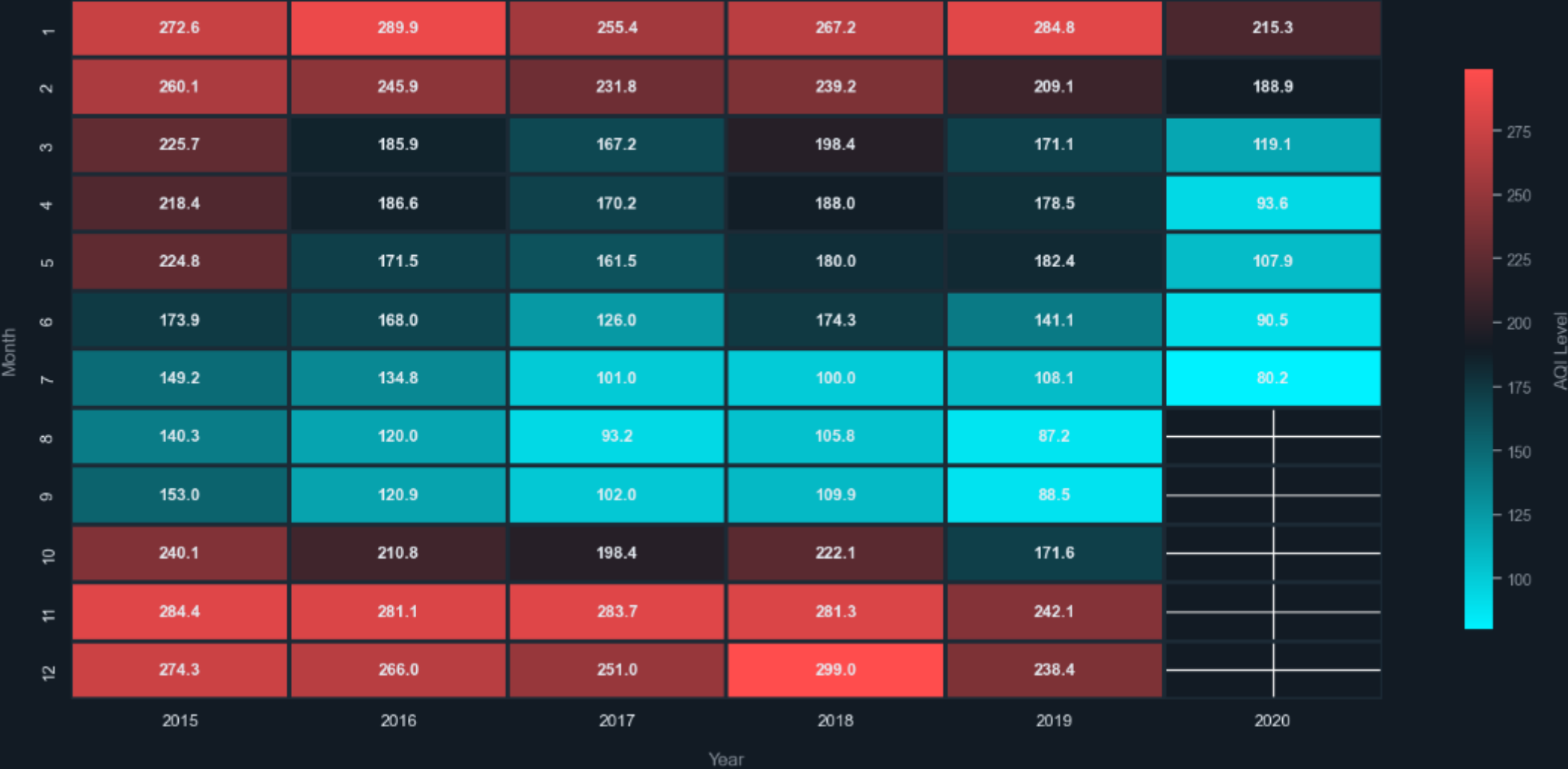
Data aggregation method

- Grouping of AQI and emission data points on a monthly and city basis and building the mean value
- Filtering noise and keeping enough data points for appropriate visualization of the correlation

Exploratory Data Analysis

Monthly AQI change over the observed period

SEASONAL AQI DYNAMICS: 2015 - 2020



Exploratory Data Analysis

Visualization of monthly AQI change over the observed period

Interpretation of monthly AQI

A serious driver for high AQI values is biomass burning. Especially in autumn and winter months there is a large-scale crop burning in agriculture fields which is regarded as low-cost alternative to mechanical tilling. In some states increased crop burning in April and May as well.

Sources on biomass burning: cropburning.in; Kaggle

SEASONAL AQI TRENDS BY YEAR (2015 - 2020)



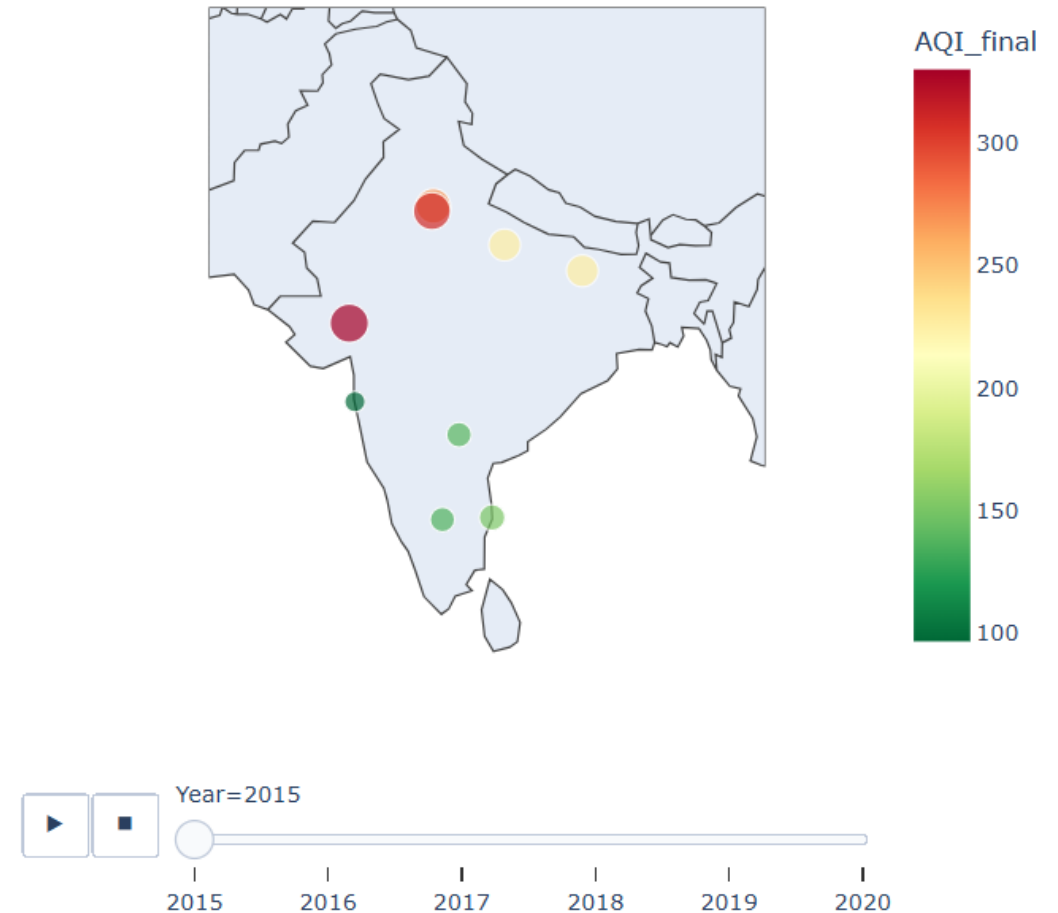
Exploratory Data Analysis

Dynamic visualization of yearly AQI on the map of India for relevant cities in the period 2015-2020

Observations and data interpretation

1. AQI generally lower in South of India compared to North of India
 - Assumption for the unequal distribution - difference in the temperature between North and South and high demand for heating during winter in the North
2. The highest yearly AQI score (626,48) - in Ahmedabad during 2018
 - Assumption – specific industry, increased crop burning, near location to Pakistan
 - In each year Ahmedabad shows the highest scores
 - Significant increase in the AQI score in Ahmedabad from 2017 to 2018 while general decrease of AQI in other relevant cities
 - In 2018, the difference in the AQI score between Ahmedabad and the second most polluted city (Delhi) is approximately 381
3. From 2018 to 2020, decrease of the AQI values for all cities (lowest in 2020)
 - Assumption – Covid-19

Average AQI distribution across Indian cities from 2015 to 2020



Machine Learning Models: SARIMA

Model choice, methodology, parameter choice and evaluation metrics

Model choice

Advantage of SARIMA Model – Special consideration of seasonality

Methodology

1. Breaking AQI values on a monthly basis
2. Choosing appropriate model parameters for the forecast: order = (p, d, q) and seasonal order (P, D, Q, s) from ACF/PACF Plots and Adfuller
3. Using TimeSeriesSplit to ensure splitting of time-ordered data
4. Fitting the model with train data to calculate evaluation metrics (Number of splits - 3 for best case with the lowest errors, 5 for worst-case with the highest errors. Number of splits below 3 and above 5 inappropriate)
5. Fitting the final model with full AQI dataset to forecast for the missing months until end of 2020

Parameter choice

Best order: (1, 1, 1)

Best seasonal order (1, 1, 1, 12)

Adfuller

ADF Statistic: 0.6776757763702986

p-value: 0.9893816580422997

The data is not stationary, we need d=1

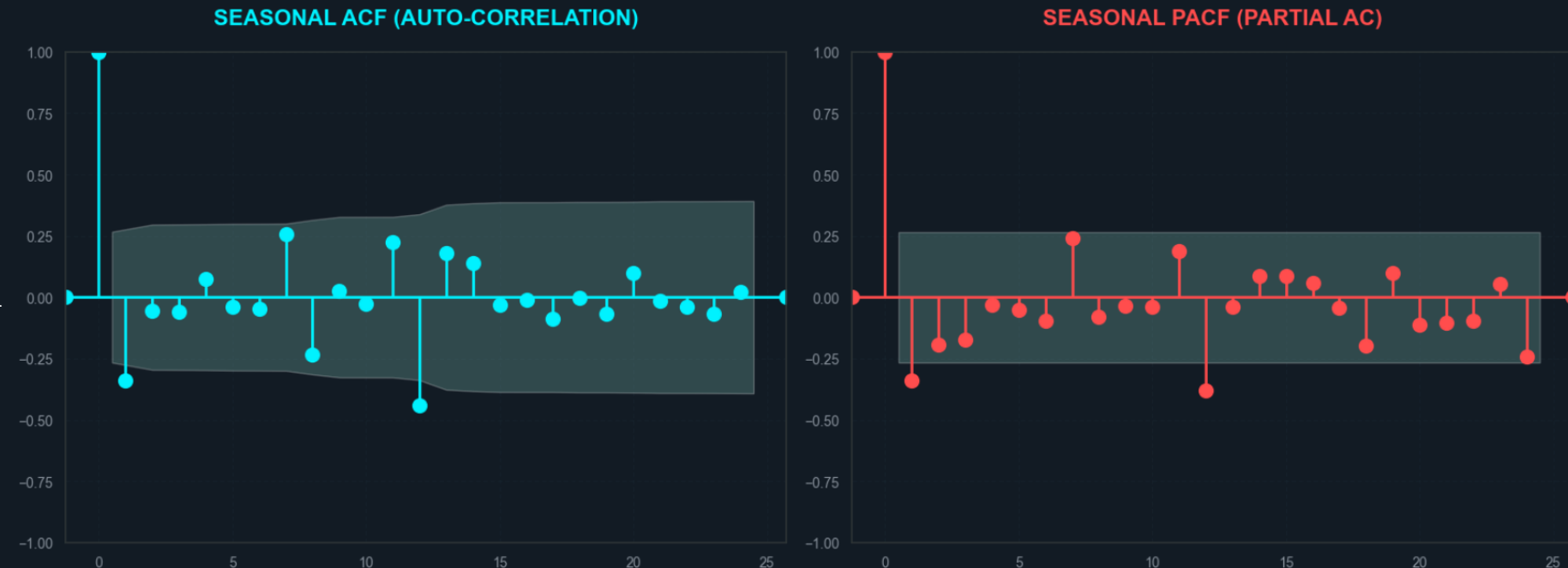
TimeSeriesSplit Validation Results

(in best case with 3 splits):

Average MSE: 1548.56

Average MAE: 31.42

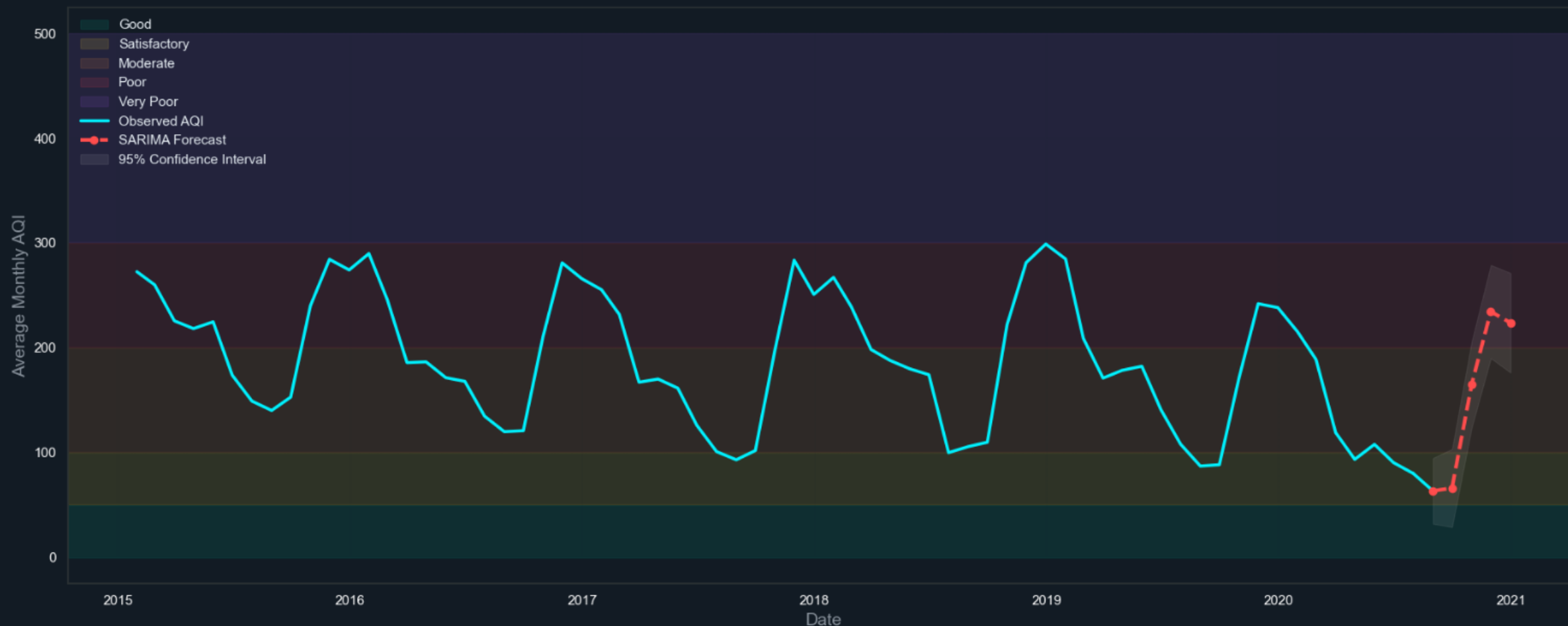
Average R^2 : 0.484



Machine Learning Models: SARIMA

AQI Forecasting through SARIMA Model

SARIMA AQI FORECAST: HISTORICAL AND PREDICTED VALUES



Machine Learning Models: LightGBM

Model training and validation

Methodology: Regime-Shift Robustness

- Training (2015 - 2019):** We trained on stable, pre-pandemic data to learn the fundamental physics of seasonal and diurnal emission cycles.
- Validation (Jan - Jun 2020):** We specifically validated on the COVID-19 lockdown era. This served as a stress test to ensure the model could adapt to the structural break caused by sudden drops in human activity.
- Forecasting (Jul - Dec 2020):** The final prediction combines the historical seasonal inertia (winter smog expectations) with the specific context of the "new normal."

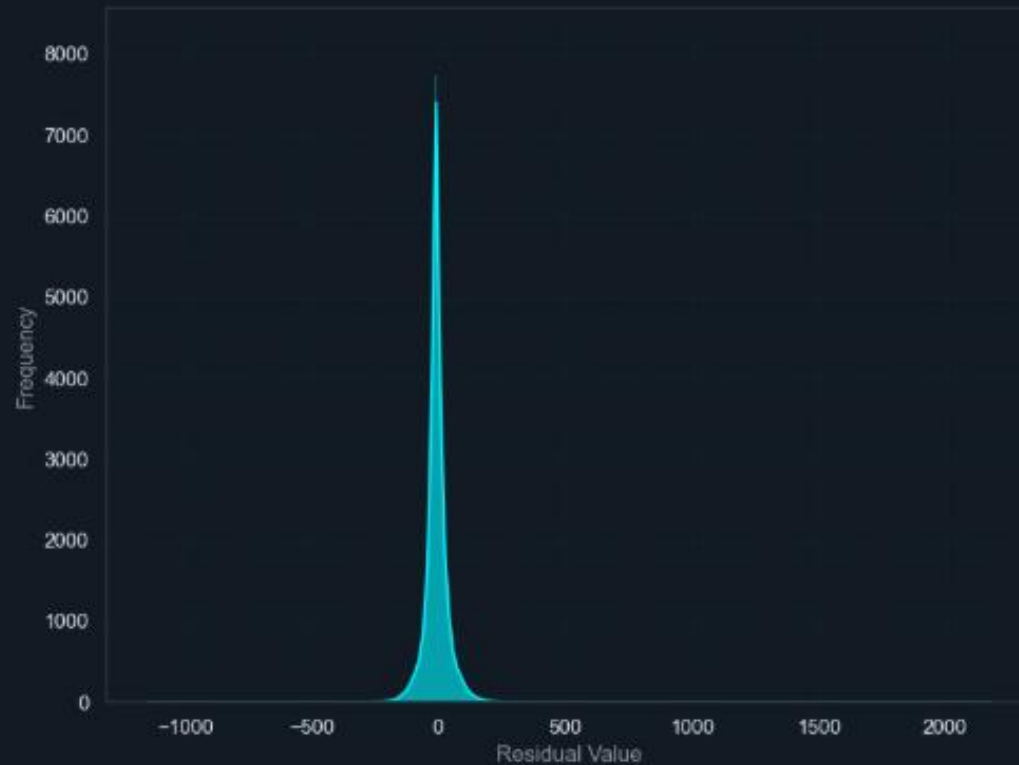
Metric	Train (2015 – 2019)	Validation (Jan – Jun 2020)
RMSE (Rooted Mean Squared Error)	66,83	49,56
MSE (Mean Squared Error)	4466,82	2456,10
MAE (Mean Absolute Error)	36,06	30,44
R-squared	0,78	0,72



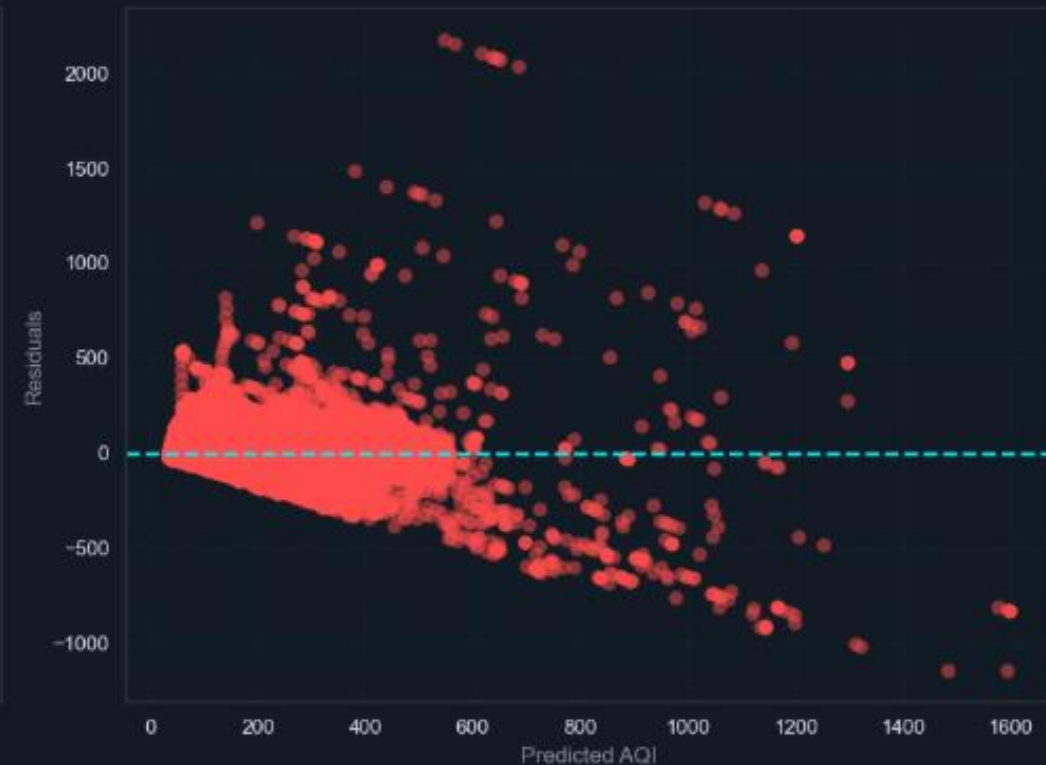
Machine Learning Models: LightGBM

Residual Analysis

DISTRIBUTION OF RESIDUALS



RESIDUALS VS PREDICTED VALUES



1. *Distribution of Residuals (Histogram)*: The residual distribution is well-centered around zero. This indicates that the LightGBM model is unbiased on average. In general the distribution slightly right-skewed, meaning that model slightly overestimates the values, but in general model is quite precise.

2. *Residuals vs. Predicted Values (Scatter Plot)*: The scatter plot reveals that the residuals are not perfectly random (homoscedastic). Instead, we observe distinct heteroscedasticity.

Machine Learning Models: LightGBM

Feature importance

The main feature in the predicting model is the AQI values lagged for 24 hours, this feature mainly reduces the error term for the predicting model.

Also mean value for 24 hour and lagged values for 1 week (168 days) appear to be quite important for the prediction. Less contribute time variables such month, day, time, year.

Recursive Forecasting Strategy

Since the standard batch prediction (e.g., `model.predict(X_test)`) is impossible for long-term forecasting because our model relies on Lag Features. For a date in December 2020, the "value from 24 hours ago" is unknown - it hasn't happened yet.

Therefore we implement an Iterative Feedback Loop:

- 1 Step:** Generate features for the current specific hour (t).
- 2 Predict:** Use the model to estimate AQI for t
- 3 Update:** Write this prediction back into the dataset. It effectively becomes the "history" used to generate lags for future steps (t+1, t+2, ...).

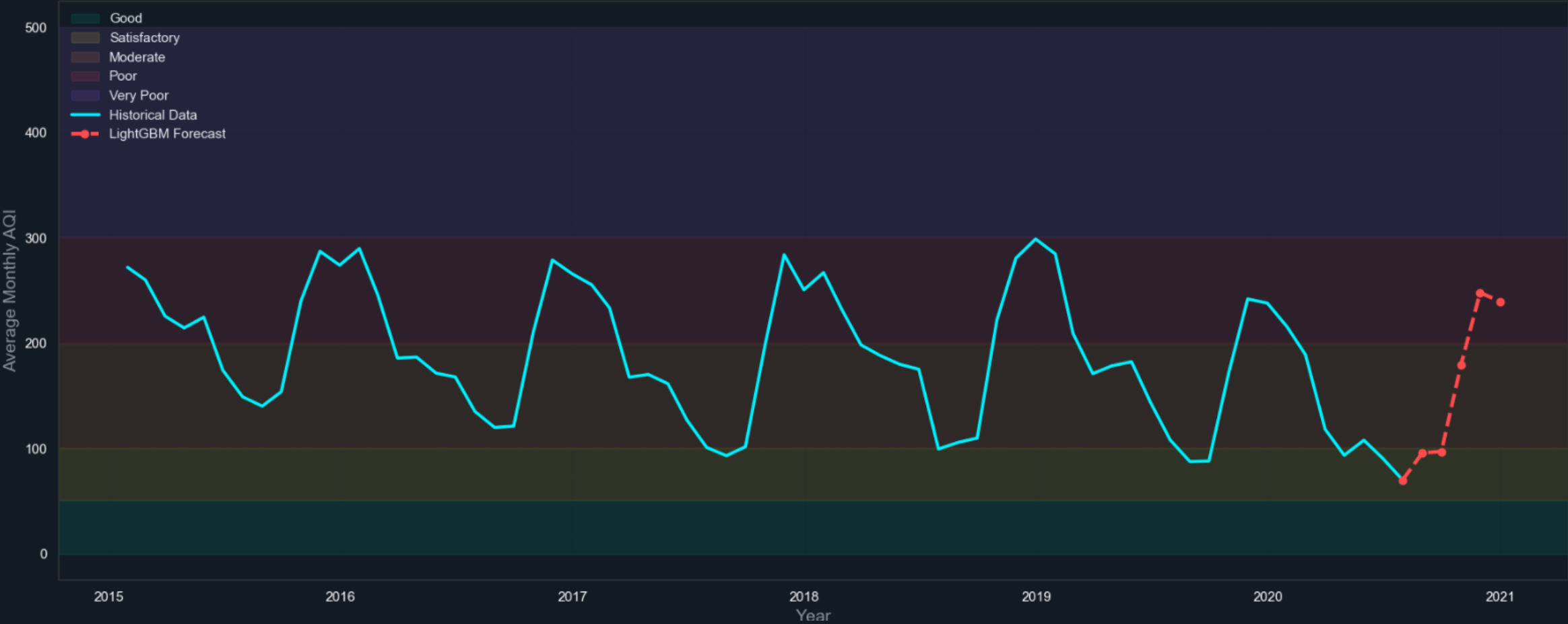
TOP 15 FEATURE IMPORTANCE (LIGHTGBM GAIN)



Machine Learning Models: LightGBM

Forecasting results

LONG-TERM AQI DYNAMICS: LIGHTGBM FORECAST



Machine Learning Models: LSTM

Model training and validation

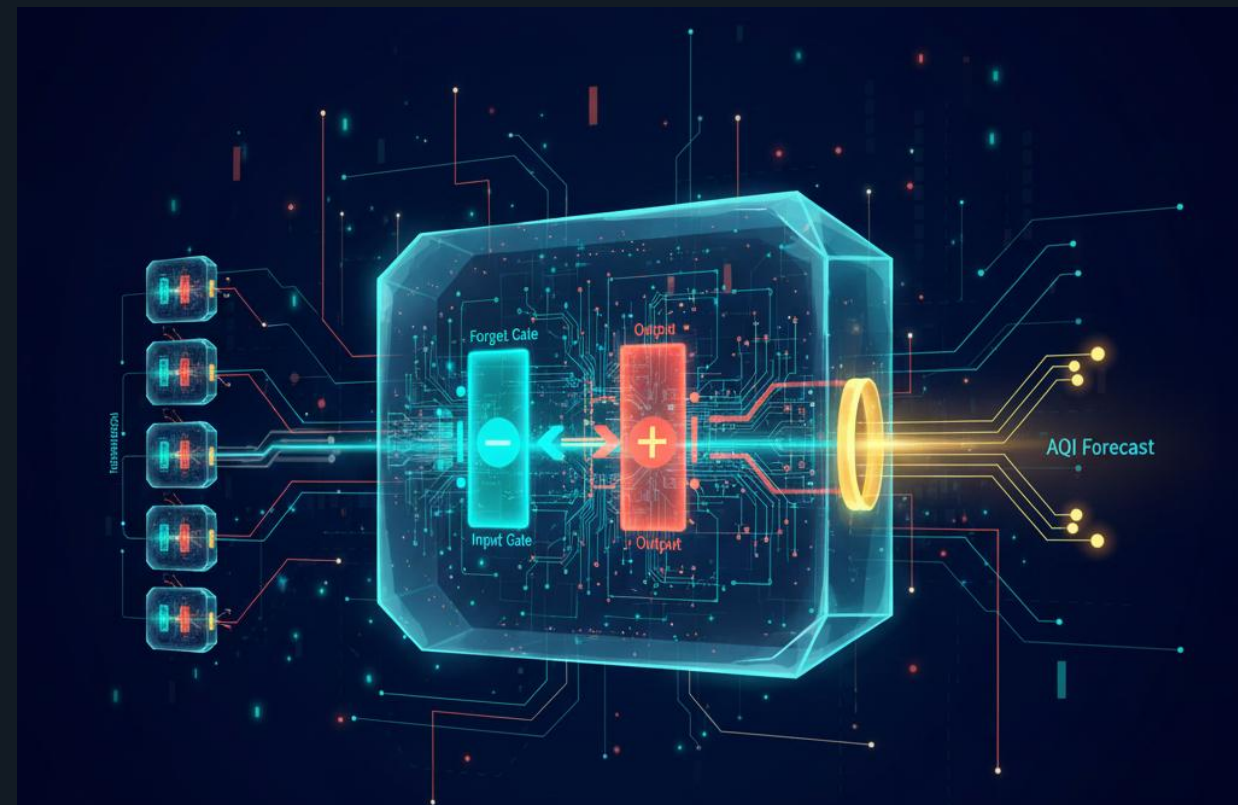
Methodology:

- Main features: seaonsality (sin_month, cos_month) historical (AQI)
- Model specifications:

Unlike standard models, LSTMs use a "cell state" to carry information across long sequences (look_back 365)

Capturing Non-Linear Relationships

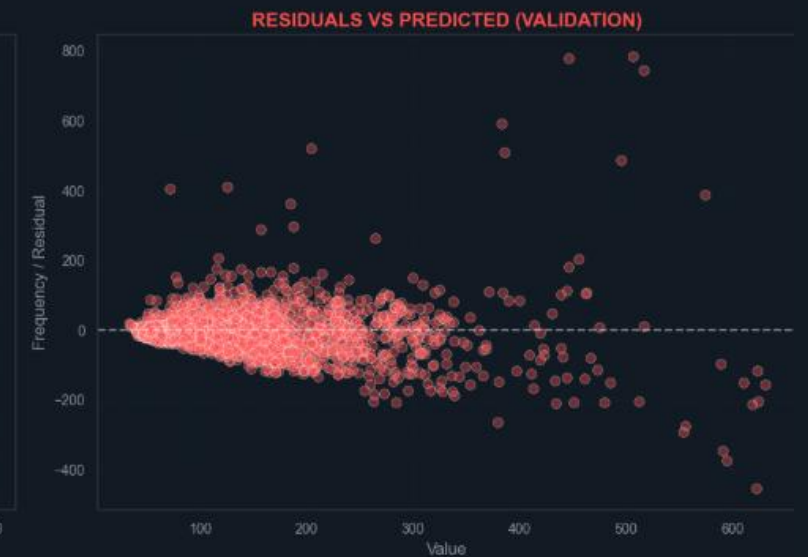
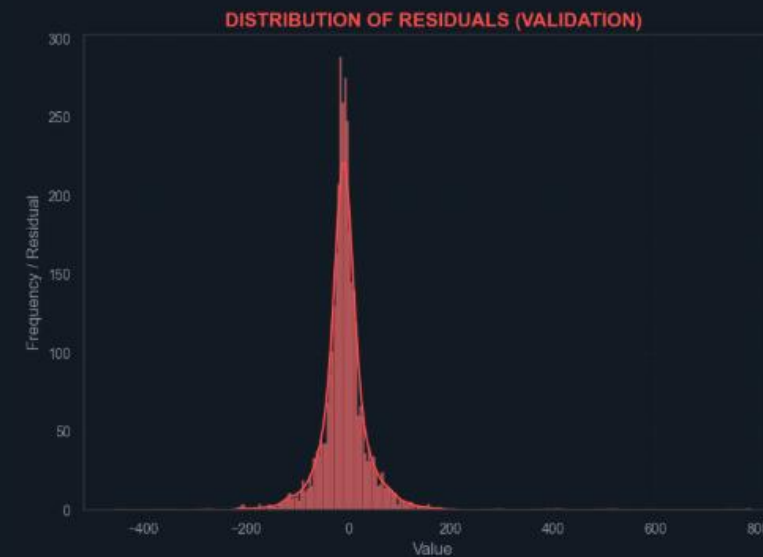
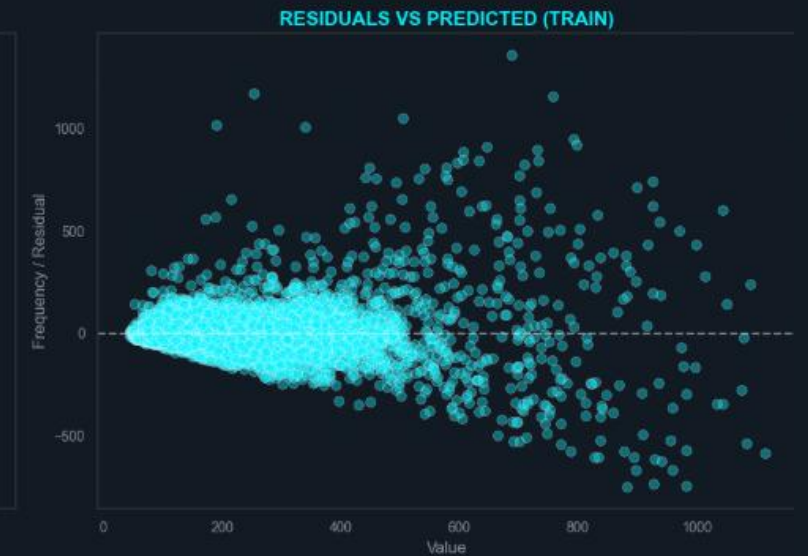
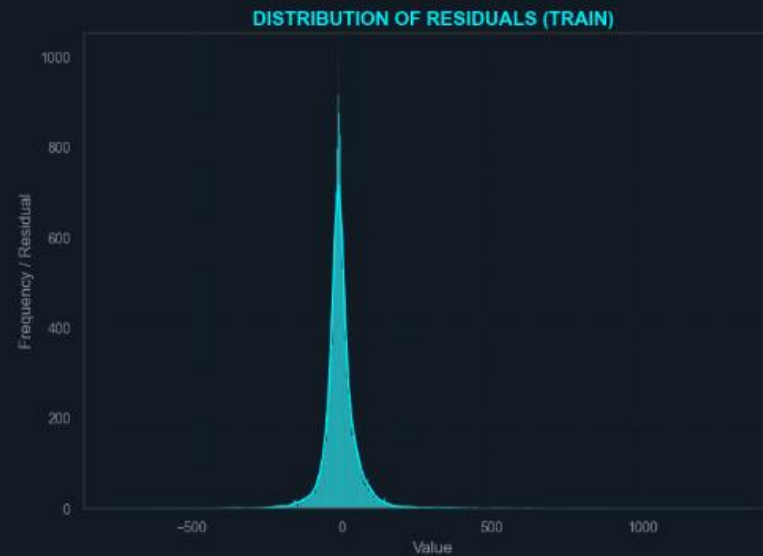
Metric	Train (2015 – 2019)	Validation (Jan – Jun 2020)
RMSE (Rooted Mean Squared Error)	84,35	56,53
MSE (Mean Squared Error)	7114,27	3195.08
MAE (Mean Absolute Error)	44,24	33.80
R-squared	0,68	0,60



Machine Learning Models: LSTM

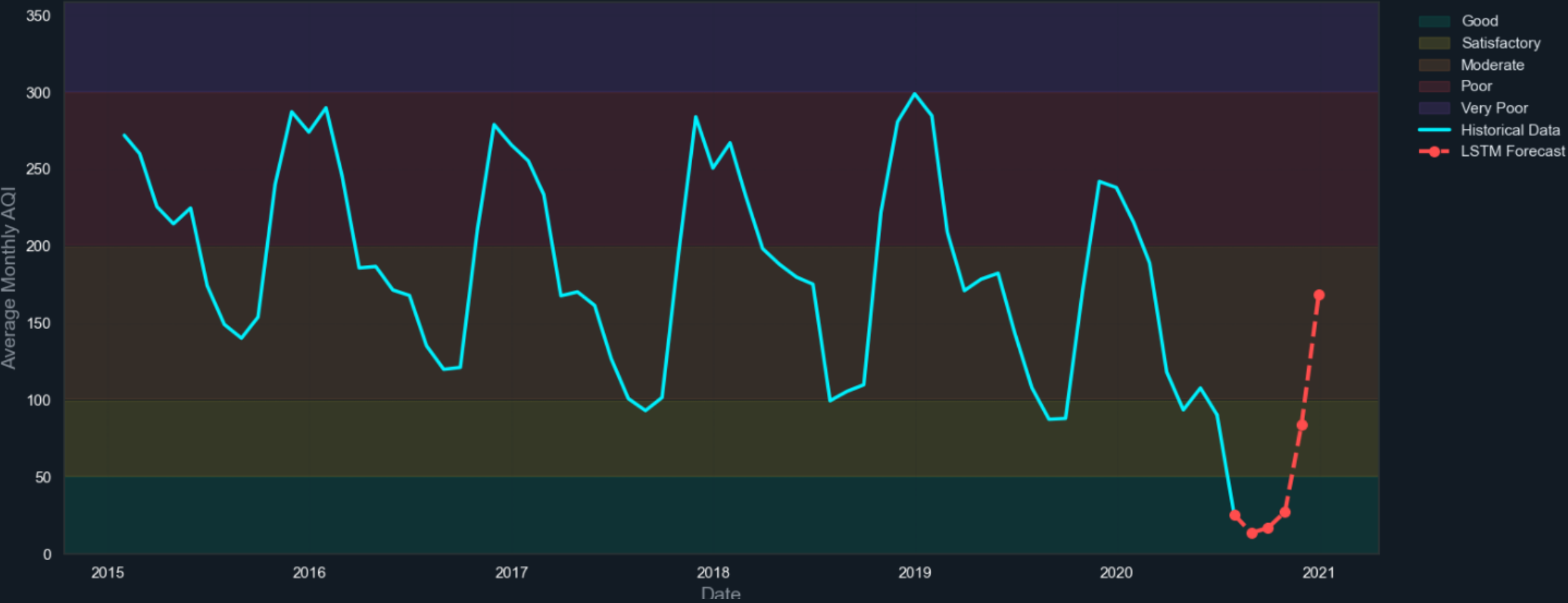
1. *Distribution of Residuals (Histogram)*: The residual distribution are positively scewed. This indicates that the LSTM model is underestimating the predicted AQI value

2. *Residuals vs. Predicted Values (Scatter Plot)*: The scatter plot reveals that the residuals are not perfectly random (homoscedastic). Instead, we observe distinct heteroscedasticity.



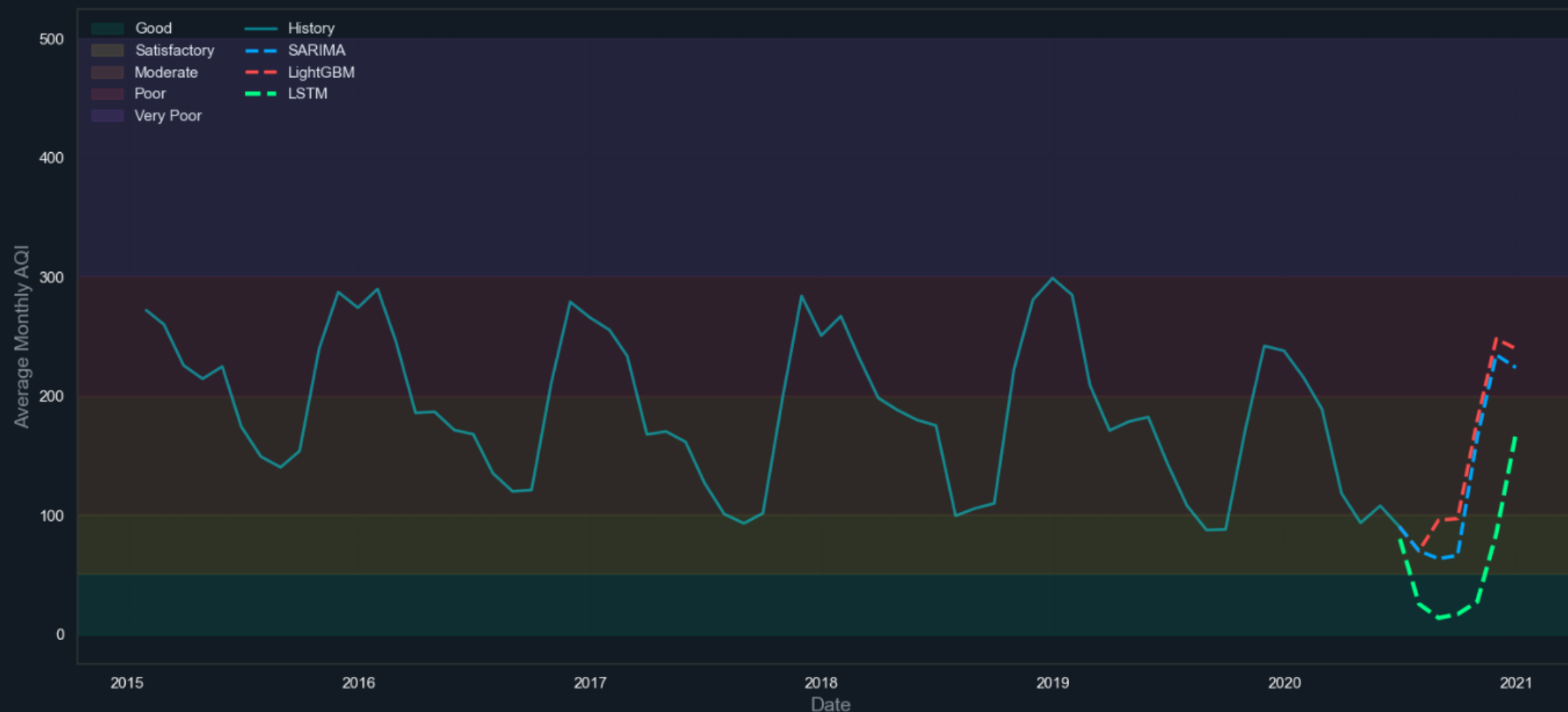
Machine Learning Models: LSTM

LONG-TERM AQI DYNAMICS: LSTM DEEP LEARNING FORECAST



Model Comparison: Samira vs LightGBM vs LSTM

FINAL MODEL COMPARISON: SARIMA vs LIGHTGBM vs LSTM



1. Patterns: strong overlapping of Sarima and LightGBM in the months from September and October
2. Divergence: Divergence in July August and November December

Clustering

This clustering analysis identifies unique 'chemical signatures' for different regions. By grouping stations based on pollutant composition rather than just total AQI, we can recommend data-driven policies tailored to the specific emission sources of each cluster.

Feature Engineering

The following metrics will be calculated for the each pollutant from the dataset:

- **Intensity ({Pol}_Median):** Represents the typical background pollution level at the station
- **Peaks ({Pol}_P95):** Represents hazardous pollution episodes.
- **Cleanest State ({Pol}_P05):** Represents the baseline pollution level when local sources are inactive or ventilation is high.
- **Pollution Amplitude ({Pol}_Range_P95_P05):** The absolute difference between the peak events and the cleanest state.
- **Volatility ({Pol}_CV):** The Coefficient of Variation. It measures how unstable the pollution source is.
- **Daily Cycle ({Pol}_DayNight_Ratio):** Compares pollution levels during active hours versus sleeping hours.
- **Weekly Cycle ({Pol}_Weekend_Ratio):** Measures the reduction in pollution when human economic activity (traffic, offices, industry) slows down.
- **Annual Cycle ({Pol}_WinterSummer_Ratio):** Captures seasonal dependency, primarily serving as a proxy for heating emissions (winter peaks) versus photochemical smog or dust (summer peaks).

Final Dataset Ready. Shape: (110, 88)					
	Benzene_CV	Benzene_DayNight_Ratio	Benzene_Median	Benzene_P05	Benzene_P95
AP001	10.426311	1.342539	0.10	0.0	0.950
AP005	0.843542	1.040221	3.35	0.0	10.258
AS001	5.483758	1.385683	0.23	0.0	6.540
BR005	1.309592	1.072803	1.54	0.0	10.656
BR006	1.227756	0.913135	1.31	0.0	7.656

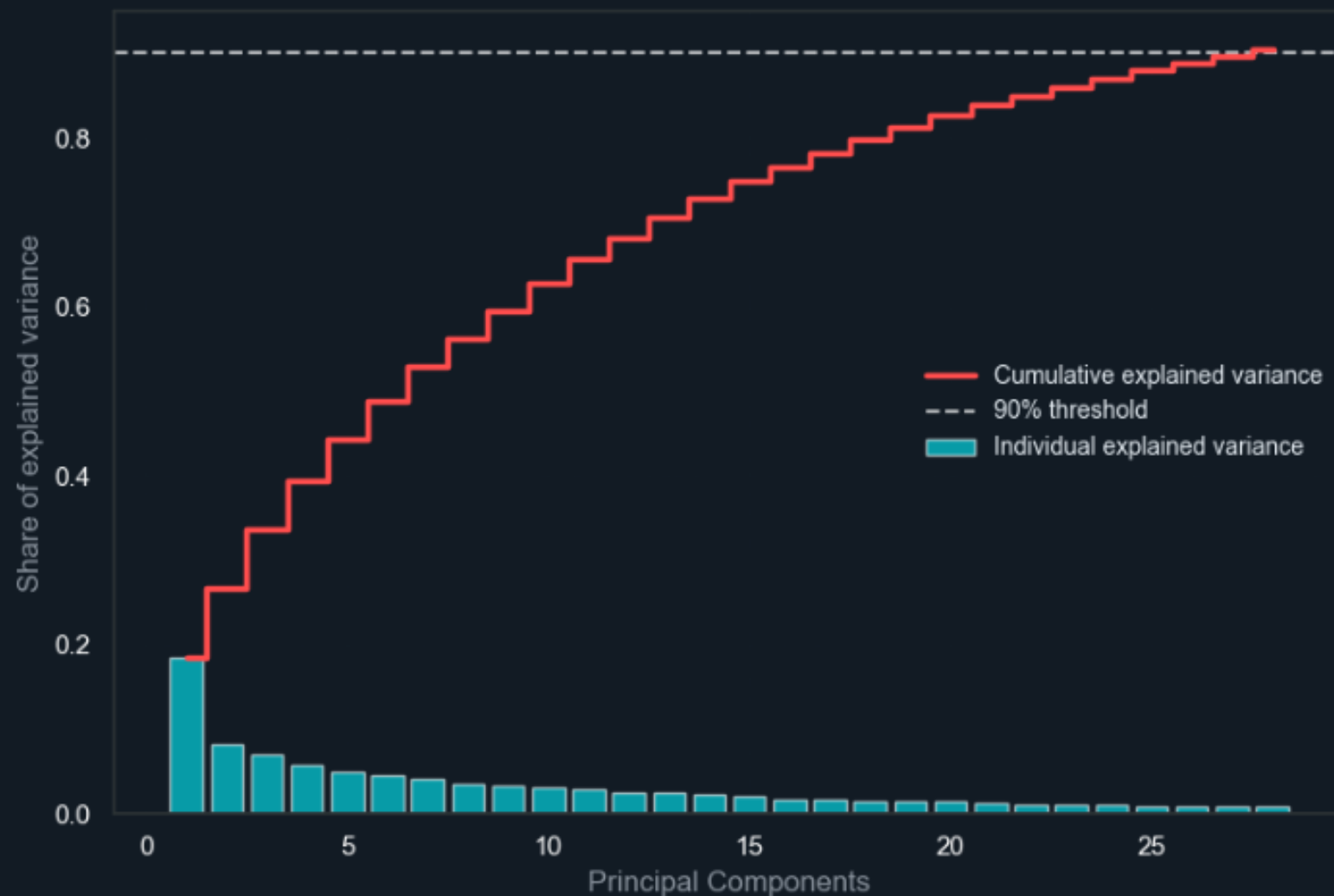
Clustering

PCA reducing dataset dimensionality

We have created dataset aggregated to station level and containing **88 features**.

This high number of variables might negatively affect the clustering result. Therefore, we need to reduce dataset dimensionality by applying PCA method.

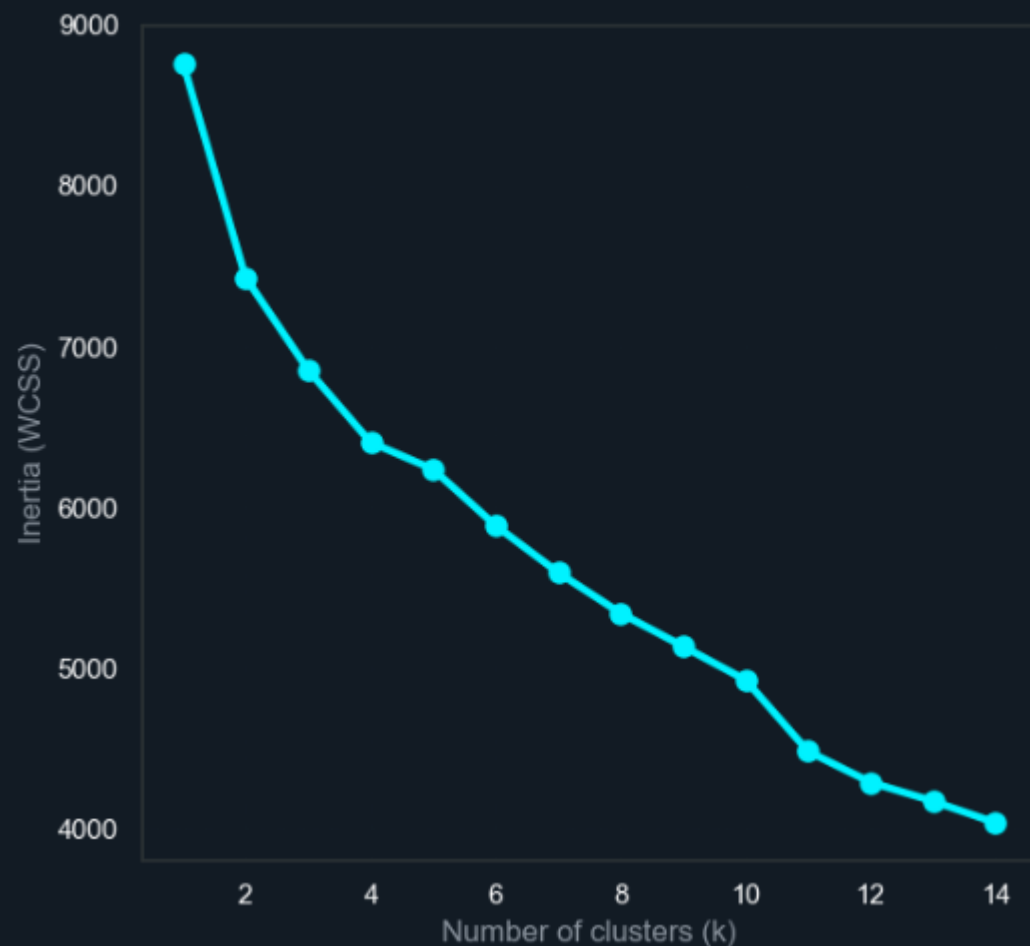
PCA SCREE PLOT: VARIANCE EXPLAINED



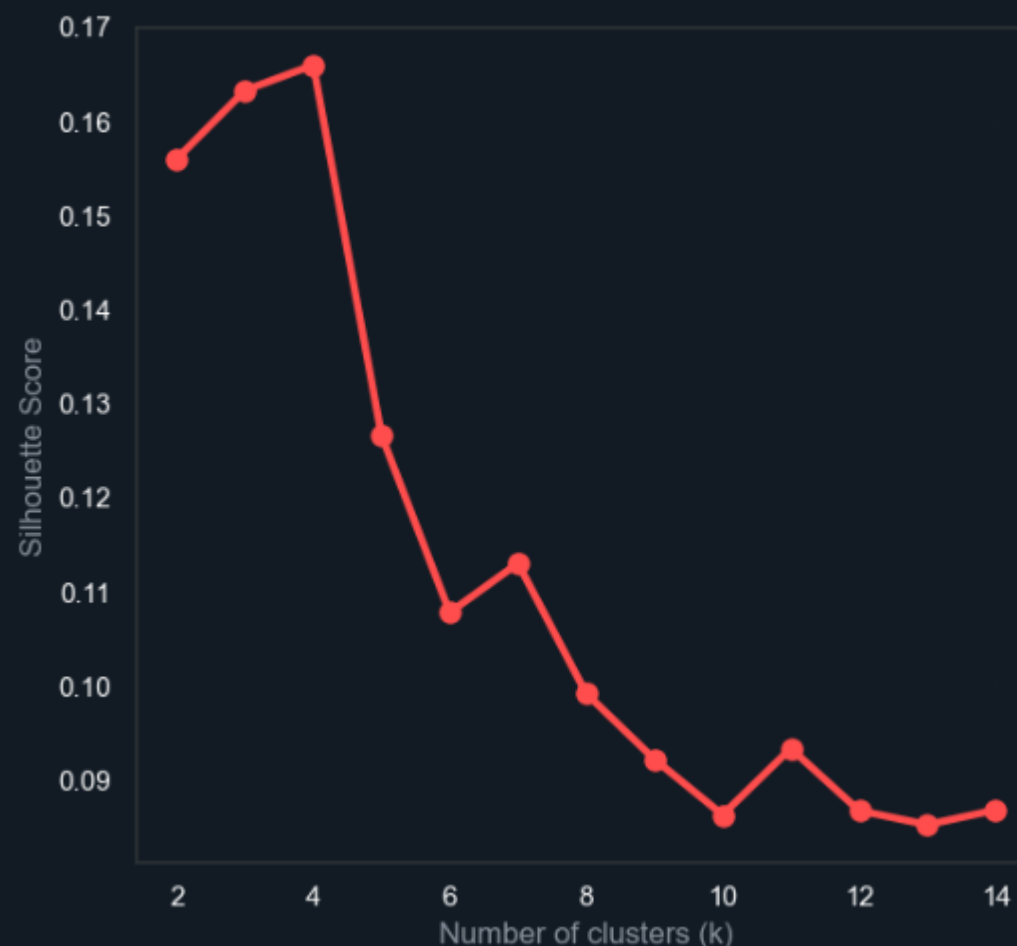
Clustering

Defining number of clusters

THE ELBOW METHOD: OPTIMAL K SELECTION



SILHOUETTE ANALYSIS: CLUSTER QUALITY EVALUATION



Clustering

Kmeans clustering

After reducing dimensionality using PCA method we have reduced number of features from 88 to 28 Principal components. However, 28 PC is still high dimensionality for clustering methods that might lead to inaccurate clustering. Therefore, not to lose information we Normalize Principal components values.

Under Normalization we understand L2-regularization. Mathematically, Normalization (L2) scales all vectors to a unit length. This transformation ensures that K-Means groups stations based on the composition of pollutants rather than just the total volume of emissions.

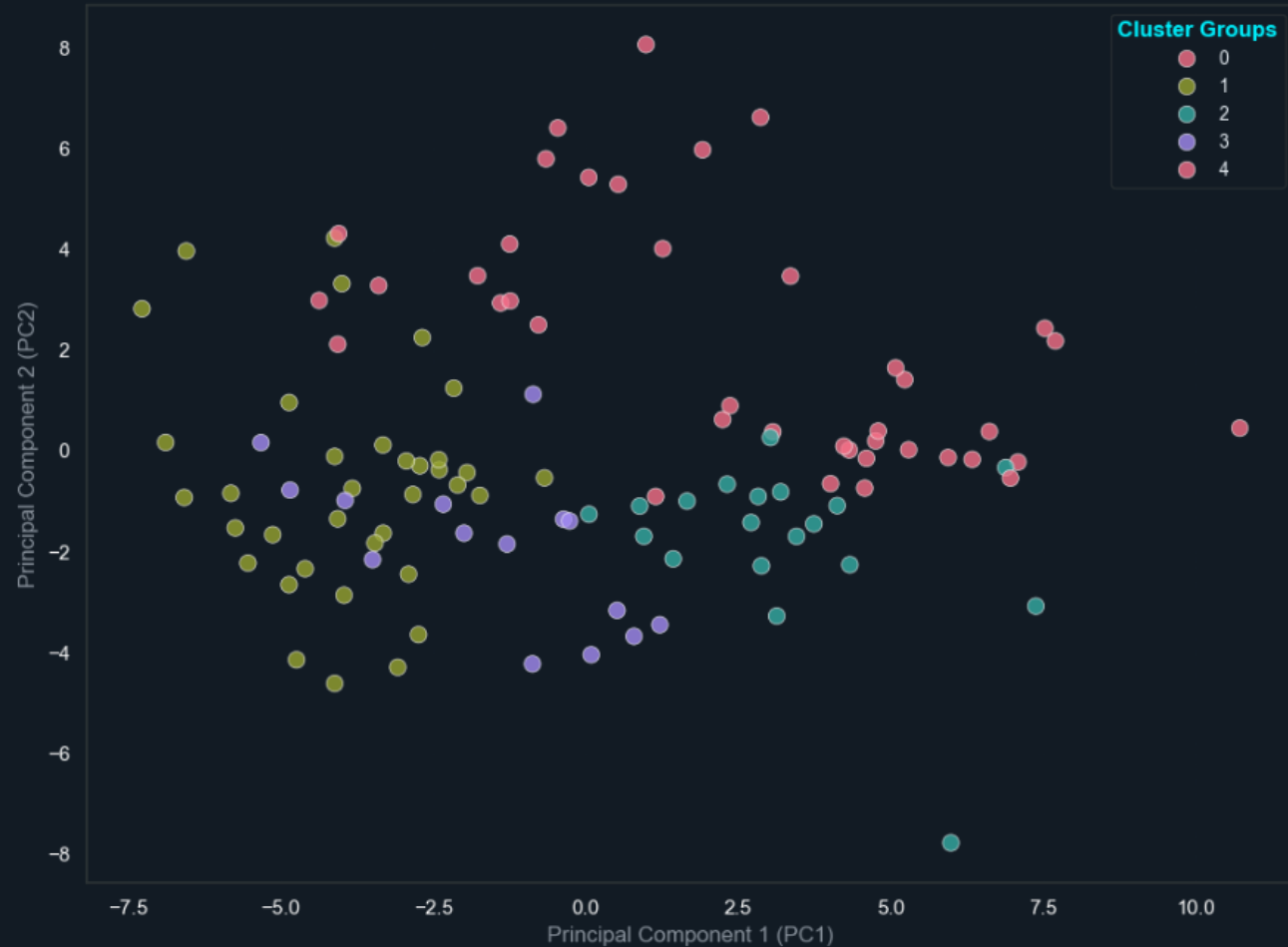
```
# Normalizing PC
transformer = Normalizer()
X_norm = transformer.fit_transform(X_pca)

# Creating, training and predicting clustering model
kmeans = KMeans(n_clusters=5, random_state=42)

labels = kmeans.fit_predict(X_norm)

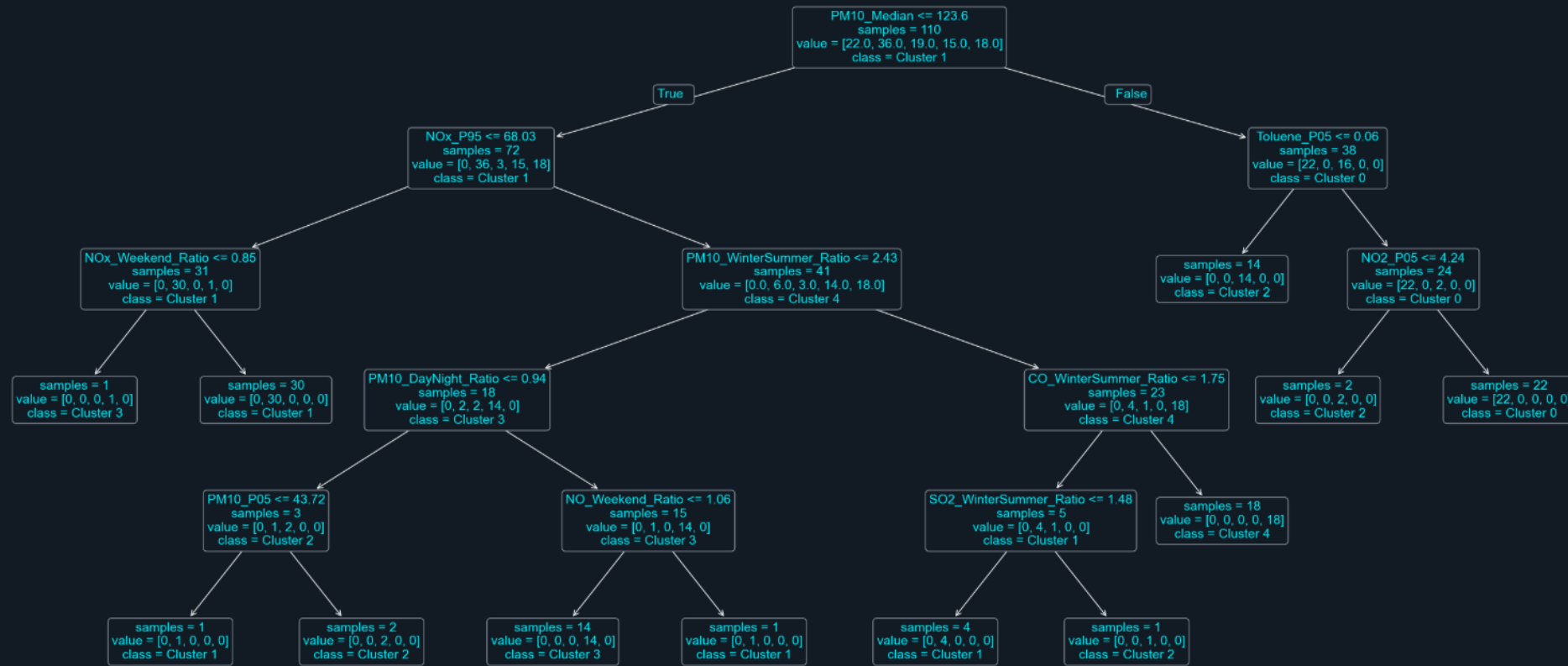
df_analysis = station_profile_final.copy()
df_analysis['Cluster'] = labels
```

GEOMETRY OF CLUSTERS: PC1 vs PC2 PROJECTION



Clustering: Surrogate Modeling - Decision Tree

DECISION TREE: CLUSTERING LOGIC (EXPLAINED)

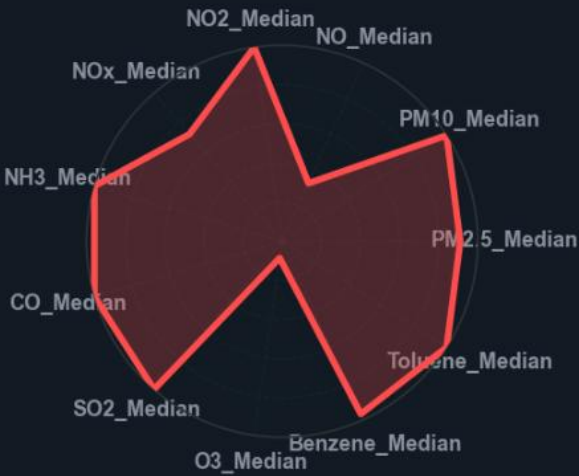


Key drivers for clustering:

- **Dominant Feature:** PM10 is the №1 factor distinguishing station profiles.
- **Gaseous Composition:** After PM10, specific gases NOx, Toluene act as the fine-tuning variables to separate similar clusters.
- The model confirms that pollution profiles are **structural and seasonal**, not just random intensity fluctuations.

Clustering: Clusters description

0: Toxic/Industrial



1: Clean



2: Dust/Power



3: Ozone



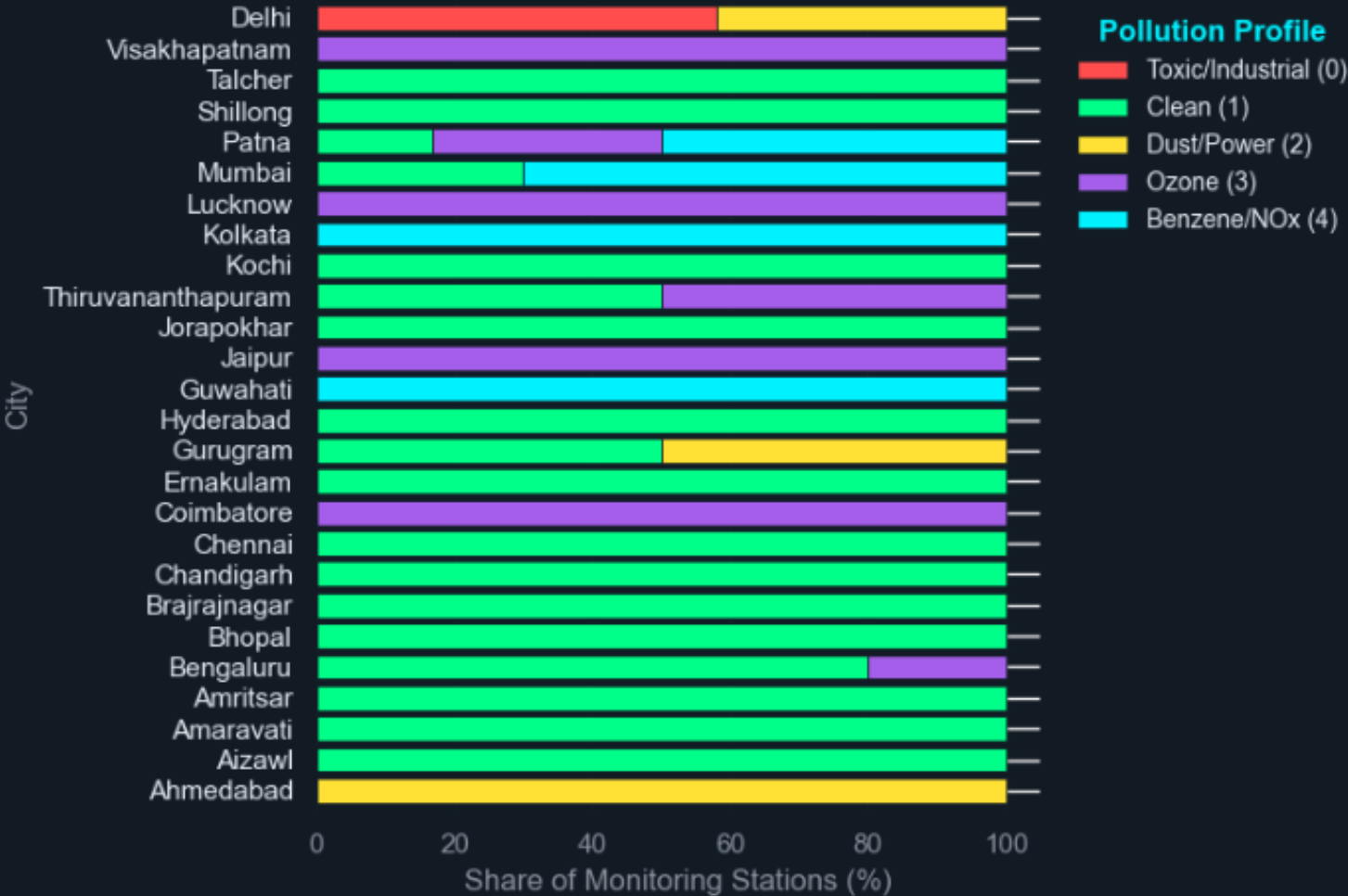
4: Petrochem/Benzene



Clustering: Description of pollution structure in cities

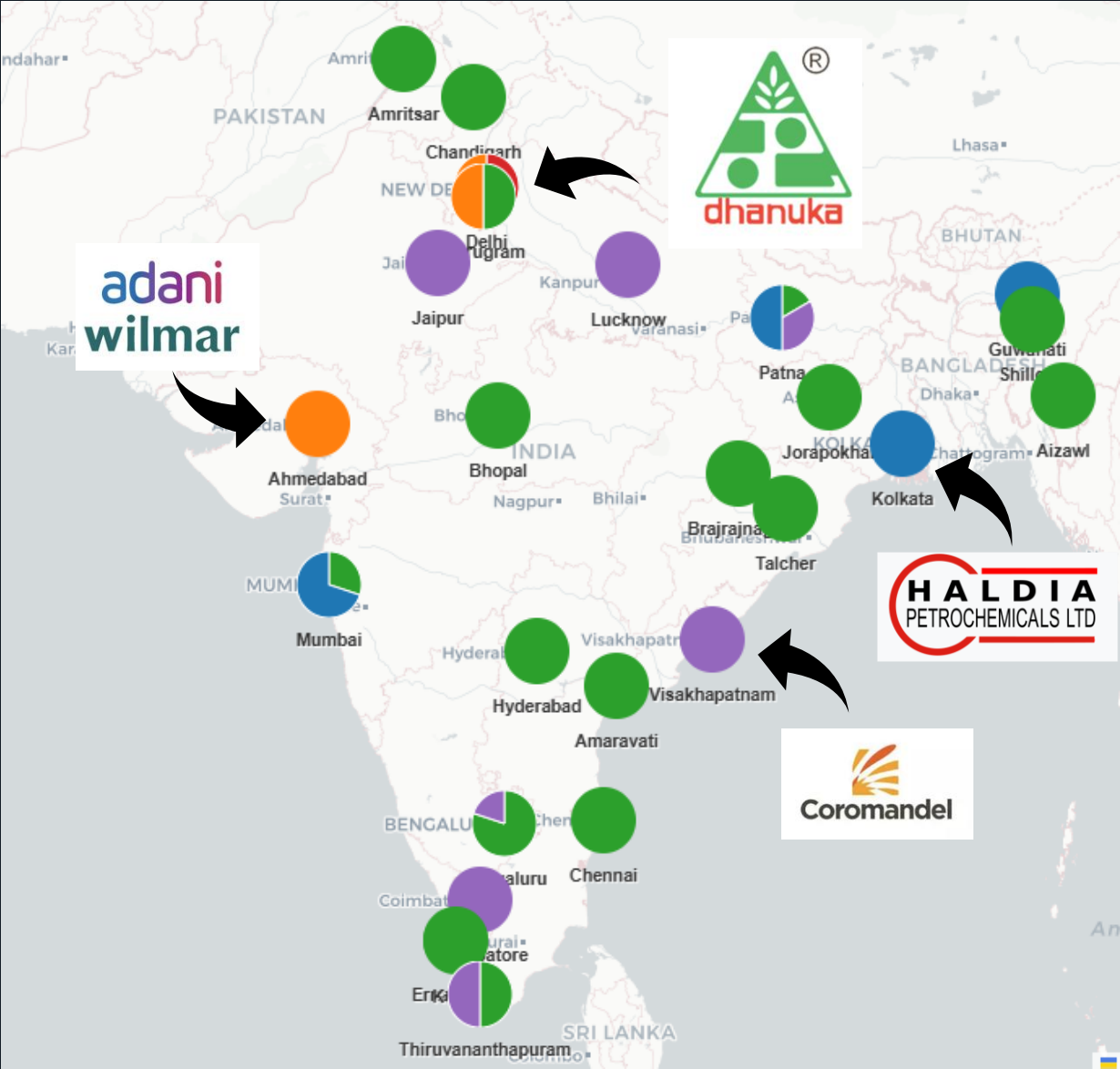
Most Pollution by Industry (Sector count)	Worst affected City	Main Pollutants	Most Dangerous Industry (Highest Risk)	Max PI
Chemicals and Explosives (9)	Dehli	Benzene, Toluene (HAPs & Heavy Metals)	Cement Plants (with co-processing)	100.0
Chemicals and Explosives (19)	Visakhapatnam	NH ₃ , O ₃ (potential) – Halogens/Acids	Synthetic Fibres (Petrochemical)	100.0
Food and Beverage Processing (27)	Ahmedabad	NO, NO ₂ , CO, SO ₂ , PM _{2.5} , PM ₁₀	Mining of major minerals	99.4
Chemicals and Explosives (6)	Kolkata / Guwahati	Xylene, Benzene (Volatile Organics)	Ammonia production (pyrolysis)	87.3

POLLUTION STRUCTURE BY CITY: CLUSTER DISTRIBUTION (%)



Matching: pollution mix with largest producer

City	Sub-Industry	Company	Key Capacity Metric
Dehli	Chemicals	Dhanuka Agritech	30,000 TPA (Fertilizer)
Visakhapatnam	Chemicals	Coromandel International	1,300,000 TPA (Fertilizer)
Ahmedabad	Food (Edible Oil)	Adani Wilmar	1,650,000 TPA (Refining)
Kolkata	Chemicals	Assam Petrochemicals	165,000 TPA (Methanol)



Information Gain Analysis

While clustering and decision trees identify what factors drive air pollution, Kriging Variance reveals where our conclusions lack reliability due to data scarcity.

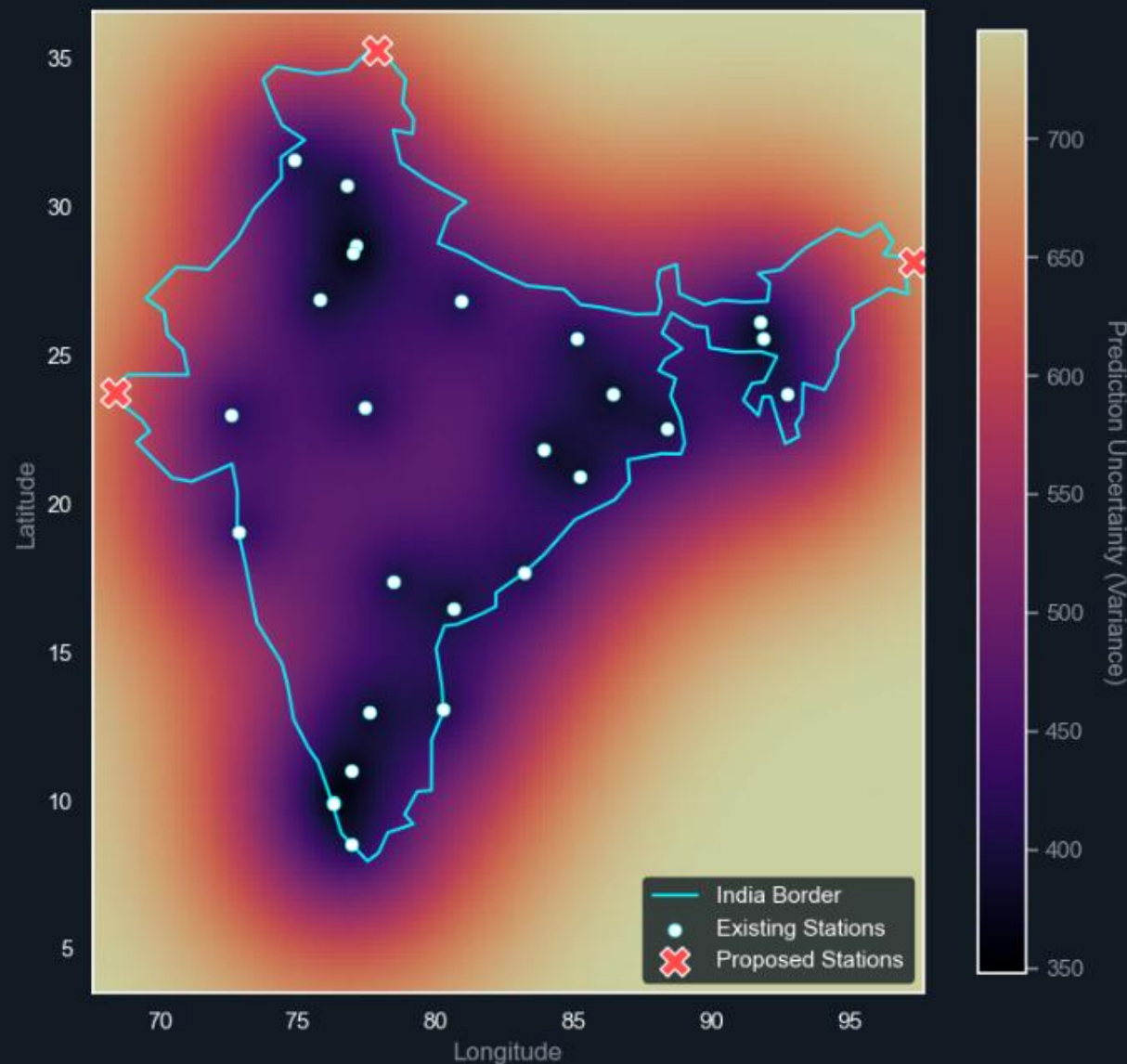
Kriging Variance

- Role: Acts as a metric of spatial uncertainty.
- Function: Mathematically pinpoints "Blind Spots" where statistical reliability is low.
- Actionable Insight: We use high-variance zones to propose specific locations for new monitoring stations.

Average normalized uncertainty: **0.27**

India's Area Coverage Index (0-1): **0.73**

STRATEGIC MONITORING GAPS: PROPOSED EXPANSION



Conclusion

Spatial analysis and trends



Multi-Factor Profiles

Delhi and Gurugram exhibit complex pollution signatures, blending Benzene/NOx with seasonal Ozone fluctuations.

Data Coverage (0.73)

High statistical uncertainty is concentrated in remote, non-urbanized regions, which likely serve as **clean environmental baselines**.

Regional Divergence

Delhi and Jaipur are improving via industrial relocation, whereas Ahmedabad remains a hotspot due to stagnant food and beverage sector emissions.

Forecast and Action



Predictive Outlook

SARIMA, LightGBM, and LSTM models show synchronous trends forecasting a pollution surge during the November–January peak season

Policy Recommendation

Implement seasonal output restrictions on the chemical sector during peak season, with a primary focus on the **Adani Wilmar facility** in Ahmedabad due to its persistent emission profile and the heightened risk of regulatory capture